CareRelay: architecture

Built for the NTU LKCMedicine challenge

Architecture, trust boundaries and key design trade-offs

16 October 2026

The care-navigation part of Challenge 1

Help an older adult act on advice already supplied.

1. **Understand the plan**
   Read it back in their own words.
2. **Identify what makes it hard**
   Choose a route and ask for consent.
3. **See the actual outcome**
   A failed request stays visible.

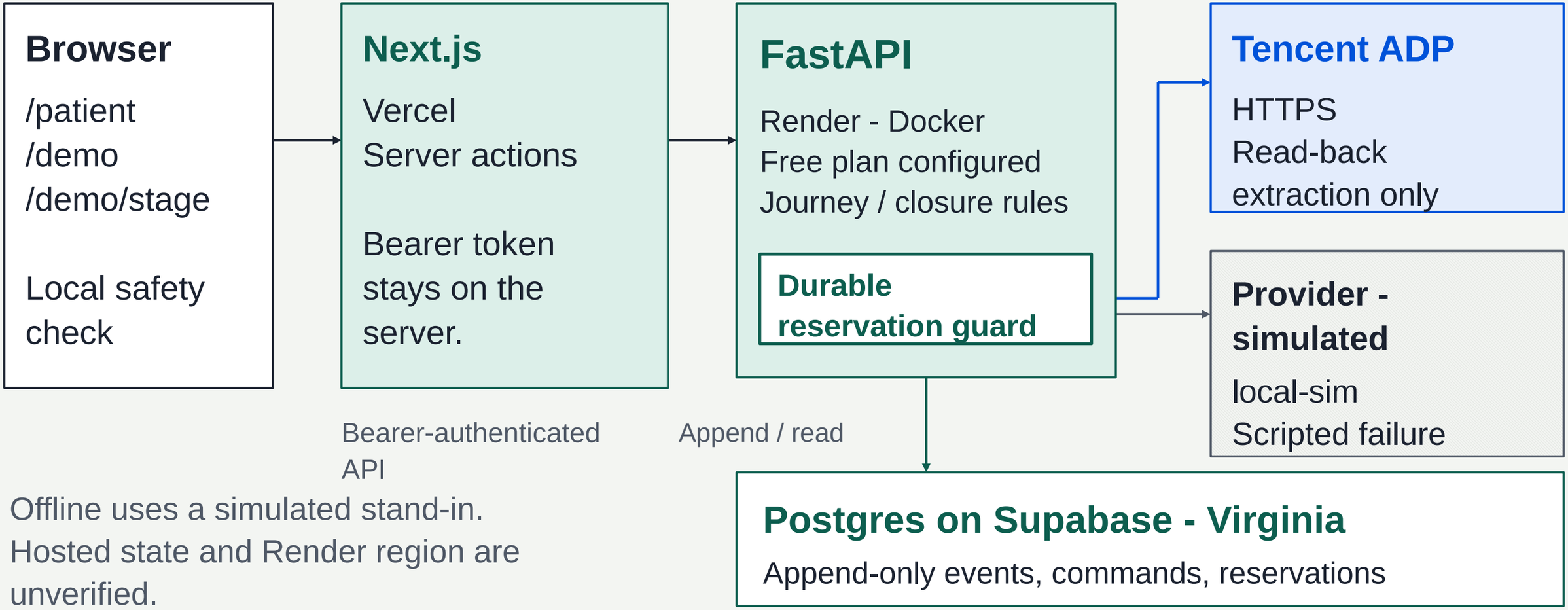
A fixed fictional plan

The plan supplies the action, deadline, who acts and fallback. The model cannot change them.

A labelled emergency screen is the only triage-like step.
No diagnosis, real booking or healthcare-service integration.

System architecture

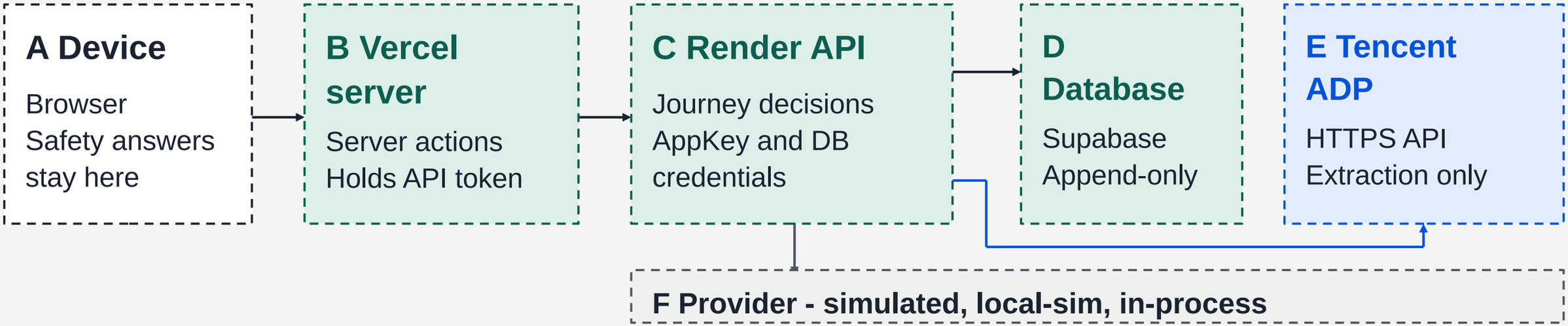
Read left to right. The guard must reserve a call before Tencent ADP is contacted.



Offline uses a simulated stand-in. Hosted state and Render region are unverified.

Trust boundaries

Dashed outlines mark trust zones. The simulated provider stays inside the API process.

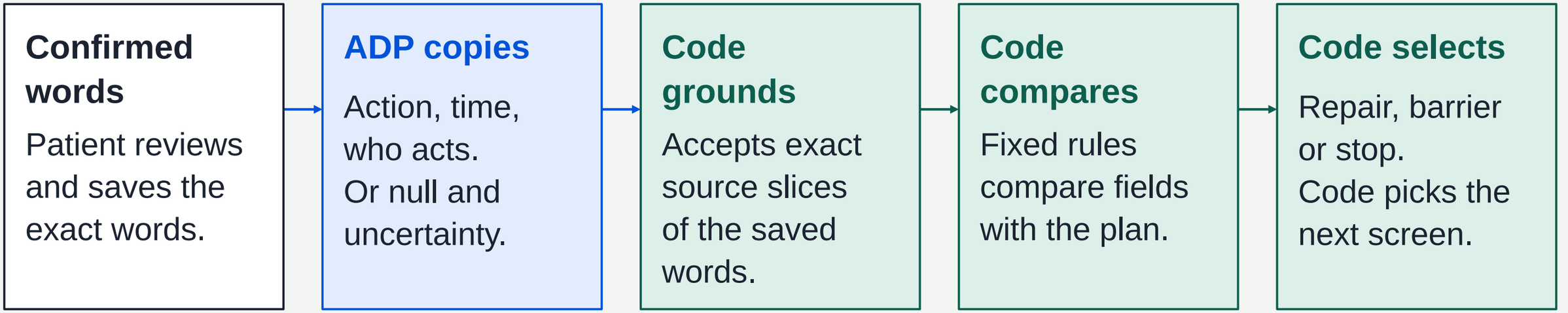


Crossing	What crosses	What does not cross
A / B	Inputs and rendered views	No server secrets. Safety answers stay on device.
B / C	Bearer token, commands and projections	No AppKey or database credentials to Vercel.
C / D	Words, events, commands and reservations	No safety answers, AppKey or raw model stream.
C / E	Confirmed words, schema, vocabulary and IDs	No expected plan, consent record or patient profile.
C / F	Permitted action and deadline, failure callback	No real provider endpoint or evidence of care.

AppKey goes only to Tencent API authentication over HTTPS. It is absent from the model prompt and browser.

The model interprets. Code decides.

Only confirmed fictional words are eligible. The stored plan remains the reference.



Strict v2 schema. 15 s whole-response cap. One call. No retry or lexical fallback.

ADP never sets the plan · sets deadline or urgency · grades understanding · books or closes anything

Offline: a labelled simulated stand-in copies phrases. The judge panel names that actor.

Safety layers in execution order

1	Before any plan **Local quick safety check**	SpO2 <90%, 30+ breaths/min, new chest pain or confusion. Any one: Get emergency help now. Call 995. No plan. Unsure or unmeasurable stays unchecked. No network or model.
2	Before ADP **Durable reservation guard**	Commit a row before the socket opens. One live call at a time. 5 per episode. 100 per Singapore day. 300 total. Invalid / timeout / transport / lost halts calls. An operator clears it.
3	After extraction **Abstention and refusal**	Clinical requests and injected instructions are refused and unscored. Uncertain fields show: We could not confirm that.
4	After a request **Truthful closure**	Execution and evidence are separate axes. A sent request is never care. A simulated failure keeps the plan, deadline and open case.

Scenario 2 example warning signs. Not clinically reviewed. A structured ADP refusal does not halt live calls.

Key design trade-offs

Decision	Why	Cost
Code decides	Explicit, testable rules.	Narrow language coverage. Rules need upkeep.
One live call at a time Pause on failure	Bounds sends across visitors and restarts.	Concurrent judges may see unavailable. An operator must clear a failure.
Free hosting tiers	Lower demo hosting cost.	Cold starts and network delay can cause timeouts.
Provider is simulated	Repeatable failure demonstration.	No real booking or healthcare connection.
Safety answers stay local	No stored safety-answer history.	No audit or judge-panel trace.
Challenge example thresholds	Labelled emergency screen.	No clinical review or validated protocol.

Latency finding: 30 to 94 SQL statements per request at about 0.2 s each caused timeouts.
The database moved to Virginia. The current Render region is [unknown].

Evidence and limits

Recorded local evidence, 16 October 2026. It does not prove hosted or clinical readiness.

2,296
automated tests passed

233
tests skipped

27 / 27
Slice 24 guard mutations detected

Supporting proof

Guard mutations expose broken guards.
Leak scans include planted controls.
A local product run saw a committed reservation before the stub socket.

Limits that remain

No clinical review. One persona.
No fairness testing across subgroups.
Postgres guard path and race unproven.
Tencent region and retention [unknown].
Hosted guard activation [unknown].

Security limit: credential-bearing Git history is disclosed. Rotation is not verified.

One demo run and the judge stage

/demo/stage frames /patient. Its read-only panel reads the ledger for the current episode.

1 Safety check

Local answers only. No judge-panel trace.

2 Plan

Panel shows the fixed plan and deadline.

3 Read-back

Panel shows saved words, phrases and comparison.

4 Difficulty

Panel shows the route code permits.

5 Consent

Permission is its own step. It is checked before dispatch.

6 Request - simulated

Panel: No reply yet. Separate callback.

7 Failure - simulated

Panel: The booking failed.
Case stays open.

8 Nurse helpline

The fallback stays visible. No call is made.

Refusal branch at read-back: clinical questions or injected instructions stay unscored.
The patient can revise the words. Guard refusal appears as Not sent to Tencent ADP.

Reload reads the same episode. The panel never creates a plan, starts a check or records a callback.